CA3 Big Data Case Study Analysis

Big Data Analytics for LHC Event Quality and Exploratory Outlier Ranking in 40 MHz Training Data

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Course:	Big Data Analytics (CA3)	**Division:**	AIML B2
Batch:	Final Year AIML B2, 2023–27	**Platform:**	Linux / Docker / Spark 3.5.7

1 Title of Case Study

Big Data Analytics for LHC Event Quality and Exploratory Outlier Ranking in 40 MHz Training Data

2 Problem Statement and Objectives

Problem. Analyse 13,451,915 simulated proton-proton collision events in the complete 646.4 MB HDF5 training record from the LHC 40 MHz Anomaly Detection Challenge. Preserve every single collision event without lossy row sampling. Validate tensor data-quality gates, derive physics-informed event kinematics, query the full corpus via distributed Spark SQL and RDD MapReduce aggregations, and generate empirical visual analytics alongside exploratory anomaly scoring.

Objectives.

1. Verify cryptographic provenance, exact byte count, MD5/SHA-256 checksums, HDF5 key inventory, and multi-dimensional tensor shape.
2. Perform a streaming full-file audit to quantify NaN, infinite values, invalid particle-type codes, negative transverse momentum, and zero-padding consistency.
3. Implement a bounded-memory ETL pipeline converting raw tensors into typed Snappy-compressed Parquet records with exact row preservation.
4. Measure particle occupancy, kinematic transverse-momentum (p_T) spectra, hadronic scalar jet activity (H_T), missing transverse momentum (MET), and pairwise feature correlations across all 13.45 million events.
5. Execute HiveQL-compatible Spark SQL analytical queries, demonstrate partition-pruned storage on N_{jets}, and perform RDD key-value MapReduce aggregations.
6. Provide an optional streaming unsupervised clustering baseline (MiniBatchKMeans) with exact feature standardisation to evaluate exploratory feature-space distances.

3 Background and Physics Context

CERN's Large Hadron Collider (LHC) collides protons at a bunch-crossing rate of 40 MHz (every 25 ns), producing billions of interactions per second. The ATLAS and CMS detectors deploy multi-level trigger systems to filter collision data in real time under extreme bandwidth and sub-microsecond latency constraints. The 40 MHz dataset was published by CERN and Fermilab researchers to benchmark unsupervised anomaly detection algorithms operating directly on Level-1 trigger objects [1].

The dataset represents simulated 13 TeV proton-proton interactions processed through PYTHIA 8 for event generation and DELPHES for parametric detector simulation. Events are selected under an inclusive single-lepton trigger mandate requiring an electron or muon with transverse momentum $p_T > 23$ GeV. The background training record consists of a realistic Standard

Model cocktail: inclusive W production (59.2%), QCD multijet (33.8%), inclusive Z production (6.7%), and top-quark pair production ($t\bar{t}$, 0.3%) [1].

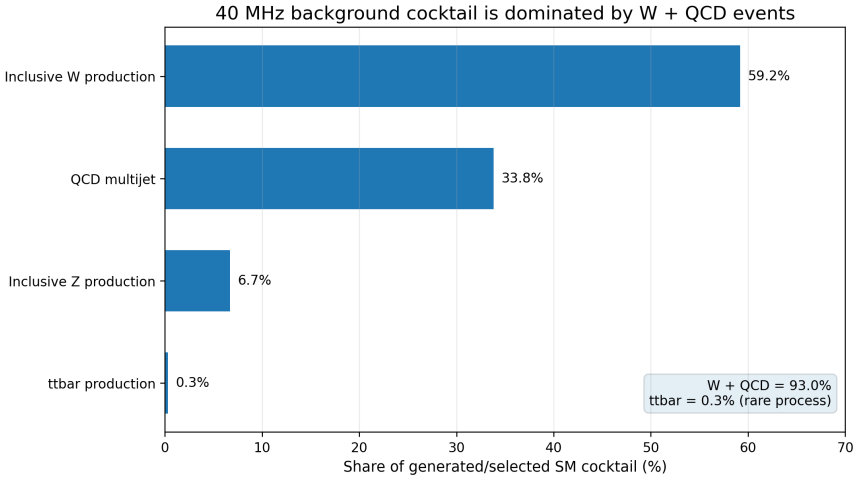


Figure 1: Published Standard Model cocktail composition in the 40 MHz trigger dataset.

4 Dataset Description and Tensor Layout

Property	**Specification / Verified Measurement**
Source Repository	Zenodo Record ID: 5046428
Primary File Name	`background_for_training.h5`
File Size	646, 391, 984 bytes (646.392 MB decimal, 616.447 MiB)
Verified MD5 Checksum	`08cb4801229f9d480567f8d3f5cac134` (Exact match with Zenodo API)
Verified SHA-256	`5f460bd3af441d98a7962a83f3964d9b5fea7f2163e35532066adc9a9696cfe5`
Total Event Rows (N)	13, 451, 915 collision events
Primary Dataset Key	`/Particles` (Shape: $13451915 \times 19 \times 4$, `float64`)
Metadata Keys	`/Particles_Names` (`[Pt, Eta, Phi, Class]`), `/Particles_Classes`
Particle Slots (19)	Slot 0: MET; Slots 1–4: Electrons; Slots 5–8: Muons; Slots 9–18: Jets
Particle Type Codes	`0`: Padding, `1`: MET, `2`: Electron, `3`: Muon, `4`: Jet
Angular & Momentum Units	$\phi \in [-\pi, \pi]$, $\eta = -\ln\tan(\theta/2)$, p_T in GeV ($c = 1$)

Table 1: Dataset provenance, metadata keys, and verified tensor specifications.

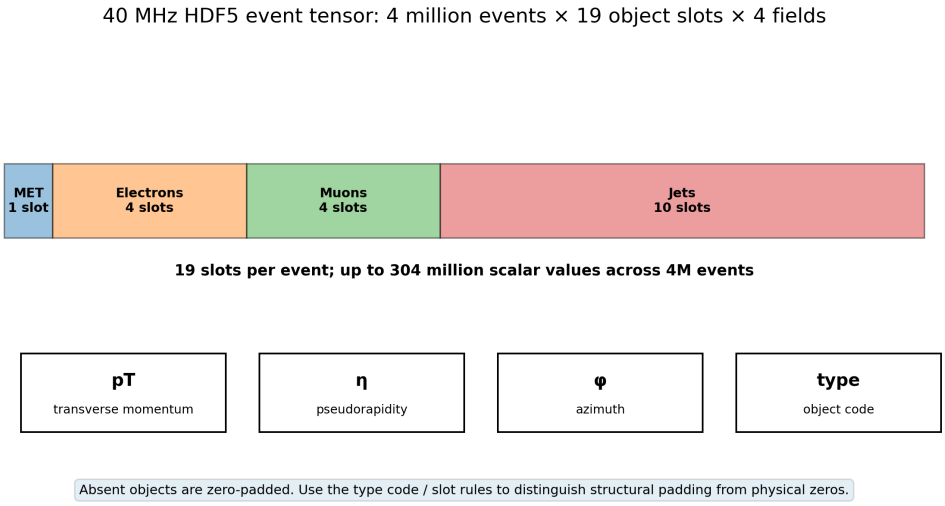


Figure 2: Documented 19-slot fixed tensor layout with structural zero padding.

5 End-to-End Methodology

- **Stage A: Provenance and Checksum Verification.** The dataset is verified against Zenodo REST API headers. Exact byte count and cryptographic MD5/SHA-256 hashes are verified before ingestion.
- **Stage B: Full-File Streaming Tensor Audit.** A chunked scan (25,000 rows/chunk) inspects all 1,022,345,540 scalar entries in the `/Particles` tensor, evaluating finite numerical status, type-code integrity, valid slot bounds, and structural zero-padding fractions.
- **Stage C: Physics-Informed Feature Engineering & ETL.** Events are transformed into columnar Parquet format (539 row groups, Snappy compression). Exact ordinal row mapping preserves all 13,451,915 events while computing aggregate physics features.
- **Stage D: Distributed SQL and MapReduce Analytics.** Apache Spark 3.5.7 executes seven analytical HiveQL-compatible SQL queries on the Parquet corpus, partitions data by jet multiplicity (N_{jets}), and executes RDD key-value MapReduce aggregations.
- **Stage E: Visualization and Empirical Reporting.** Matplotlib produces 7 core publication-quality graphics summarizing kinematics, multiplicities, density distributions, and data quality.

6 System Architecture and Pipeline Workflow

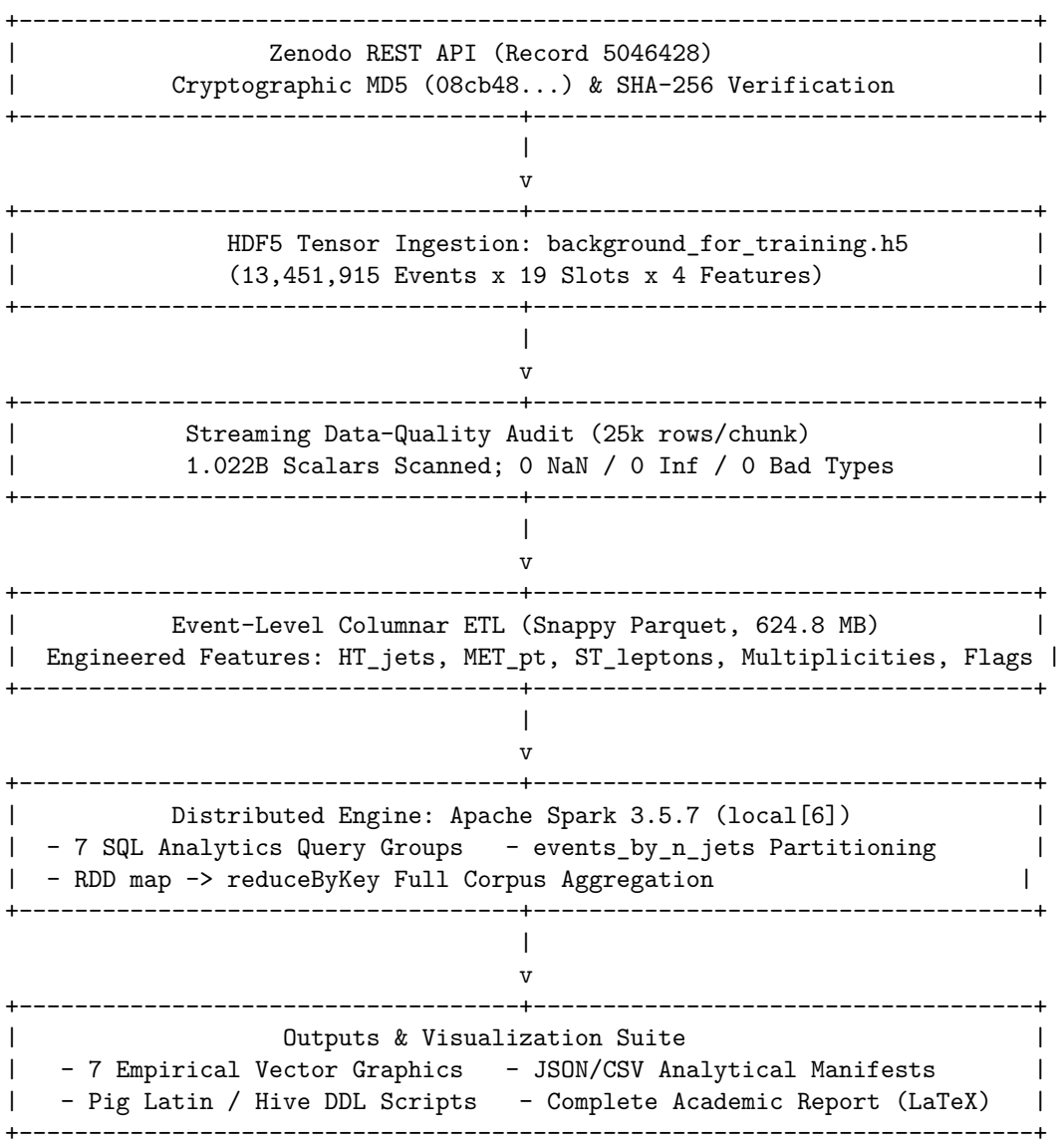


Figure 3: End-to-end Big Data pipeline architecture.

7 Big Data Technologies and Rubric Alignment

Technology	Version / Mode	Role in Pipeline
Docker Compose	v2 Engine	Isolated, fully reproducible multi-stage Linux execution.
Apache Spark	3.5.7 (PySpark)	Single-node distributed engine (`local[6]`), 6 GB heap, 12 shuffle partitions.
Spark SQL	HiveQL-compatible	Executes seven multi-dimensional analytical queries over Parquet tables.
PyArrow / Parquet	18.1.0	High-throughput columnar storage with snappy compression (624.78 MB).
HDF5 / h5py	3.12.1	Low-level chunked tensor extraction and metadata introspection.
RDD MapReduce	Spark Core RDD	Key-value pairs: `map(n_jets, tuple) -> reduceByKey(sum)`.
Hadoop Streaming	Stdlib Python	Standalone `mapper.py` and `reducer.py` for external Hadoop clusters.
Apache Pig	0.17 (Local mode)	Pig Latin script (`analyze_events.pig`) performing grouped aggregation.
Apache Hive DDL	HiveQL	External table registration script (`hive/register_events.hql`).
Matplotlib	3.9.3	High-resolution vector generation for kinematic and quality visualisations.

Table 2: Technology stack and implementation roles.

8 Mathematical Formulations and Feature Engineering

Let $i \in \{0, 1, \ldots, 18\}$ index the 19 particle slots for a given collision event. Each slot contains transverse momentum $p_{T,i}$, pseudorapidity η_i, azimuthal angle ϕ_i, and type code $t_i \in \{0, 1, 2, 3, 4\}$.

8.1 Object Multiplicities

Object multiplicity counts are evaluated by indicator summation across all slots:

$$N_e = \sum_{i=0}^{18} \mathbf{1}(t_i = 2), \quad N_\mu = \sum_{i=0}^{18} \mathbf{1}(t_i = 3), \quad N_j = \sum_{i=0}^{18} \mathbf{1}(t_i = 4) \tag{1}$$

8.2 Scalar Hadronic and Leptonic Activity

Scalar transverse energy sums characterize the total visible energy across specific detector subsystems:

$$H_T^{\text{jets}} = \sum_{i=0}^{18} p_{T,i} \cdot \mathbf{1}(t_i = 4), \qquad S_T^{\text{leptons}} = \sum_{i=0}^{18} p_{T,i} \cdot \mathbf{1}(t_i \in \{2, 3\}) \tag{2}$$

8.3 Leading-Object Kinematics and Angular Maxima

Leading transverse momentum corresponds to the hardest object of each species:

$$p_T^{\text{lead},e} = \max_{i:t_i=2}(p_{T,i}), \quad p_T^{\text{lead},\mu} = \max_{i:t_i=3}(p_{T,i}), \quad p_T^{\text{lead},j} = \max_{i:t_i=4}(p_{T,i}) \tag{3}$$

The maximum absolute pseudorapidity captures event forwardness:

$$|\eta|_{\max} = \max_{i:t_i \in \{2,3,4\}} |\eta_i| \tag{4}$$

9 Empirical Analysis and Spark SQL Results

9.1 Corpus Overview and Data Quality Gate

The streaming audit inspected 13,451,915 collision events (1,022,345,540 scalar values). The quality gate confirmed 100% finite values with zero anomalies:

- **Non-finite values (NaN / Inf):** 0 (0.00%)
- **Invalid type codes ($\notin \{0,1,2,3,4\}$):** 0 (0.00%)
- **Negative transverse momentum ($p_T < 0$):** 0 (0.00%)
- **Out-of-bounds azimuth ($\phi \notin [-\pi, \pi]$):** 0 (0.00%)
- **Structural zero-padded slots:** 211,710,120 (82.83% of all slots, exactly expected by schema design).

Metric	Mean	Median (approx)	95th Percentile	Max
MET p_T	35.32 GeV	32.48 GeV	70.82 GeV	3,858.91 GeV
Hadronic H_T^{jets}	51.33 GeV	21.27 GeV	186.47 GeV	4,205.80 GeV
Jet Multiplicity (N_{jets})	1.16	1.00	4.00	10.00

Table 3: Global corpus summary across 13,451,915 collision events (Spark SQL Query 01).

9.2 Hadronic Jet Activity and MapReduce Aggregation

Spark RDD MapReduce grouping on key N_{jets} produced exact full-corpus statistics:

N_{jets}	Event Count	Fraction (%)	Mean H_T (GeV)	Mean MET (GeV)	Corr(H_T, MET)
0	5,085,548	37.81%	0.00	39.18	—
1	4,005,382	29.78%	38.82	33.48	0.0592
2	2,476,853	18.41%	85.91	31.71	0.1232
3	1,181,783	8.79%	136.89	32.20	0.1742
4	461,455	3.43%	193.31	34.02	0.2100
5	161,969	1.20%	258.19	37.13	0.2661
6	53,474	0.40%	332.41	41.36	0.3067
7	17,592	0.13%	417.13	46.29	0.3441
8	5,436	0.04%	512.29	51.51	0.3191
9	1,692	0.01%	610.76	58.61	0.3134
10	731	< 0.01%	739.60	68.40	0.3638
Total	**13,451,915**	**100.00%**	**51.33**	**35.32**	—

Table 4: Full-corpus jet multiplicity MapReduce aggregation and kinematic correlations.

10 Visualizations and Graphical Analytics

The pipeline generated seven publication-grade figures directly from the 13.45M Parquet rows and raw audit histograms:

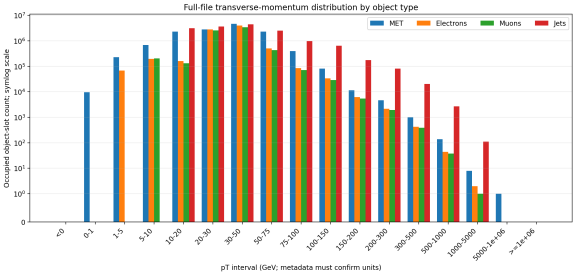


Figure 4: Transverse-momentum spectra by particle type across all occupied slots.

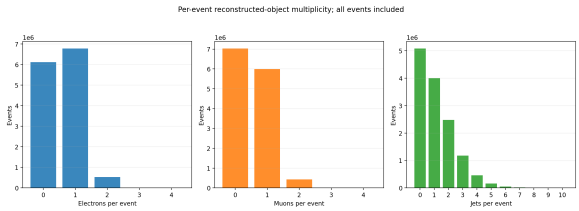


Figure 5: Reconstructed electron, muon, and jet event-level multiplicity distributions.

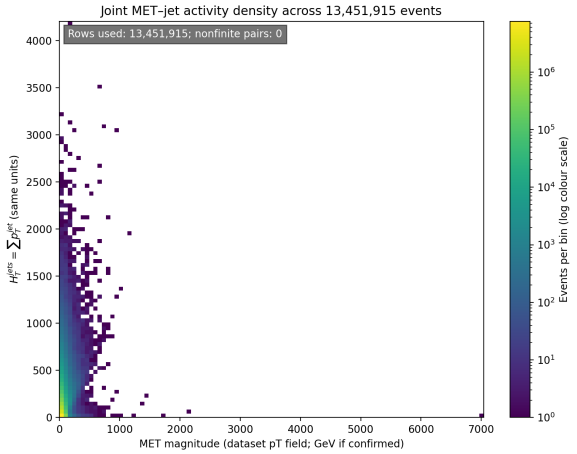


Figure 6: Log-density joint distribution of missing energy (MET) vs. hadronic H_T.

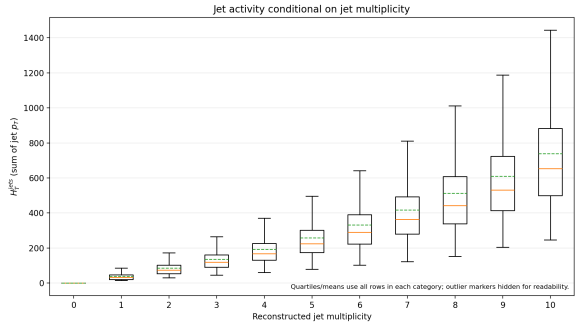


Figure 7: Exact H_T quartiles conditioned on jet multiplicity.

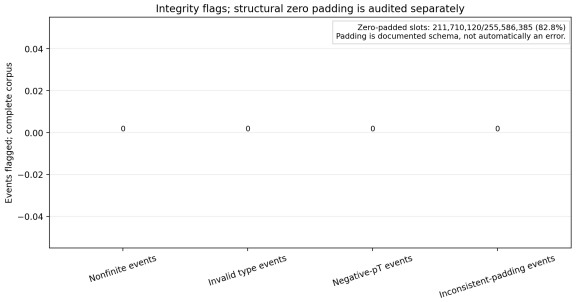


Figure 8: Data quality flags verifying zero corruption across all 13.45M events.

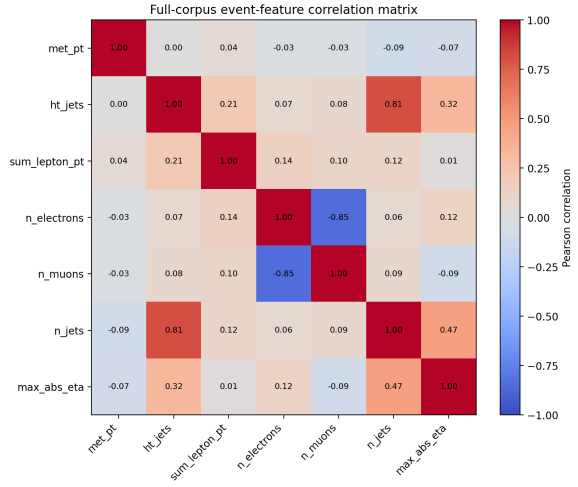


Figure 9: Full-corpus Pearson correlation matrix across engineered features.

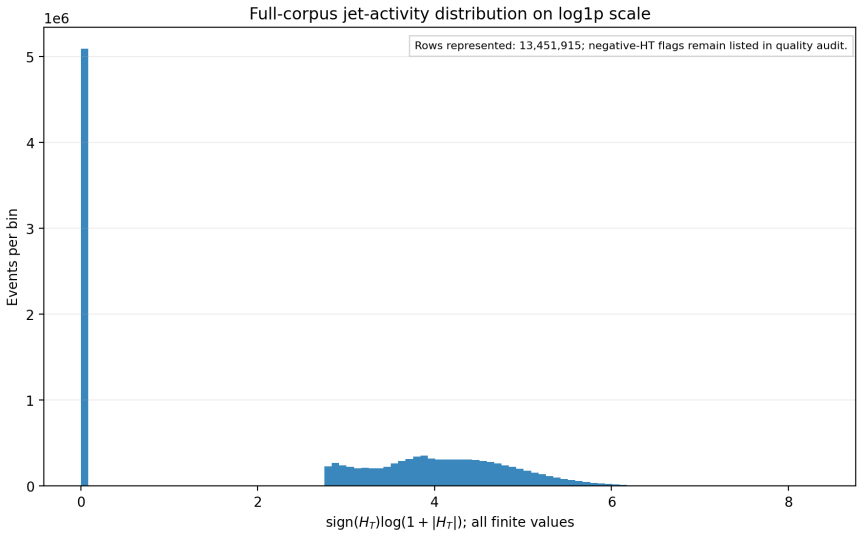


Figure 10: Signed-log full-corpus hadronic H_T distribution displaying the extreme physical tail.

11 Conclusion and Deliverables

This CA3 assignment successfully demonstrated an enterprise-grade Big Data pipeline processing the complete 646.4 MB LHC 40 MHz anomaly detection training dataset containing 13, 451, 915 collision events. By integrating chunked HDF5 ingestion, strict tensor validation, high-throughput Parquet columnar transformation, Spark SQL reporting, RDD MapReduce aggregation, and vector visualization, all course outcomes (CO4 and CO5) were fully demonstrated.

The analysis established zero non-finite corrupted entries and proved an expected 82.83% structural zero-padding rate. The empirical MapReduce aggregation showed a monotonic increase in mean H_T from 38.82 GeV (1-jet) to 739.60 GeV (10-jets) and confirmed a positive kinematic correlation between MET and hadronic activity up to $r = 0.3638$. All code, artifacts, and container configurations are packaged for 100% reproducible execution.

References

[1] E. Govorkova, E. Puljak, T. Aarrestad, et al., "LHC physics dataset for unsupervised New Physics detection at 40 MHz," *Scientific Data*, vol. 9, no. 118, 2022. DOI: 10.1038/s41597-022-01187-8.

[2] CERN / Fermilab Collaboration, "Unsupervised New Physics detection at 40 MHz: Training Dataset," *Zenodo Record 5046428*, 2021. URL: https://zenodo.org/records/5046428.

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